



**C23-C-402**

**23425**

**BOARD DIPLOMA EXAMINATION, (C-23)**

**MARCH/APRIL—2025**

**DCE – FOURTH SEMESTER EXAMINATION**

**DESIGN AND DETAILING OF R.C. STRUCTURES**

*Time : 3 hours ]*

*[ Total Marks : 80*

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**PART—A**

3×10=30

- Instructions :**
- (1) Answer **all** questions.
  - (2) Each question carries **three** marks.
  - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. State any three reasons for providing steel as reinforcement.
- 2. Compare limit state method and working stress method.
- 3. State the situations in which doubly reinforced sections are used.
- 4. Calculate development length for a steel bar of 12 mm diameter of Fe415 grade using M30 concrete.
- 5. Distinguish between one-way slab and two way slab.
- 6. State IS code provisions for providing torsion reinforcement at corners in simply supported two-way slabs.
- 7. An isolated T-beam has a flange of 1500 mm × 100 mm and web of 250 mm × 440 mm. Find the effective width of flange if the span is 5.0 m.
- 8. List any three IS code provisions for longitudinal reinforcement in design of columns.
- 9. State Rankine's formula for minimum depth of foundation.
- 10. List IC numbers of any three DACs.

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*[ Contd...*

## PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.  
(2) Each question carries **ten** marks.  
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** (a) Explain the working of clamper circuit. 5  
(b) Explain the working of Zener diode clipper with wave forms. 5
- 12.** Draw and explain the circuit of transistor type Hartley oscillators. 10
- 13.** (a) Compare the merits and demerits of RC phase shift and Wein bridge oscillators. 5  
(b) Explain Barkhausen criterion for oscillators. 5
- 14.** Explain the functional block diagram of an operational amplifier. 10
- 15.** (a) Derive the expression for voltage gain of Inverting amplifier. 5  
(b) Classify ICs based on manufacturing process. 5
- 16.** Explain the working of op-amp based astable multi-vibrator with waveforms. 10
- 17.** Explain the working of op-amp based Wein-bridge oscillator circuit. 10
- 18.** Explain D/A conversion using R-2R ladder network. 10

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